1. Trust your tire model (see how to tyre model)
    a. For this we will be trusting the NNTM
2. Gather all necessary parameters into a config
    a. See the single source of truth whenever it gets made
3. These are the values we need to get to to initiate the solver loop
    a. Front/Rear Static Tyre Loads
    b. Front/Rear Load Transfer Sensitivity
    c. Front/Rear to CG distance
    d. A Tyre Model
    e. Weight
    f. Looping values for Chassis Slip (Beta) and tyre steer (Delta) angles (With ackermann considered)
    g. Max Ay
        i. I know it doesn't sound like a lot but it takes a lot to get to these values and something could be slightly wrong at every step
4. There are three loops to contend with, the outermost loop iterates over Beta/Delta. The next loop iterates over the other. The innermost loop solves for steady state.
5. While getting into the innermost loop, we break our velocity down into its components (Vx, Vy). We account for ackerman at this point as well and split delta into delta_l and delta_r. We prime a guess for Ay, this value can be optimized for efficient solving but no matter what value is chosen we will get an answer. Set yaw rate to none or 0. And init an iteration tracker to use a as a debug tool.
6. Loop time: Base condition if Ay (new Ay) and Ay_guess (old Ay) are within a margin (some small but achievable number 0.01)
    a. Check if we’ve timed out with a maximum iteration number (optional)
        i. If yes, break, if no increment iter
    b. If the yaw rate isn't none (ie after our first time through the loop)
        i. Update Ay_guess in a damped fashion, Ay (the new Ay) * Ay_guess (previous Ay) in some ratio (70/30 seems to work well but its possible to damp it more or less)
    c. Set/Reset the yaw rate to Ay_guess / Vx
    d. Get the tyre slip angles by accounting for tyre steer, chassis slip, toe angles, the cars sideways velocity, and the cars forward velocity. (in ratio form for the sideways and forward velocities)
    e. Clap Ay based on the maximum theoretical Ay and current Ay guess (Should never become relevant to the final solution)
    f. Calculate the delta load transfer of the car on an axle by axle basis.
    g. Update current tyre loads using the static tyre loads and load transfer
    h. Plug in the tyre slip angles and normal loads into the tyre model to get Fy and Mz for each tyre.
    i. Update the Ay based on the tyre forces sum the Mz of each tyre as Mz total, return to top
7. Immediately after the convergence loop but still double nested
    a. Save the found Ay to an array

    b. Calculate and save the found Mz (yaw moment) to an array
        i. Front tyre forces perpendicular to the lever arm
        ii. Front Tyre forces parallel to the lever arm acting across the trackwidth
        iii. Rear tyre forces perpendicular to the lever arm
        iv. + Mz_total
8. Once all points have been found, pass back out the arrays to the rest of the script for graphing and KPI extraction.